

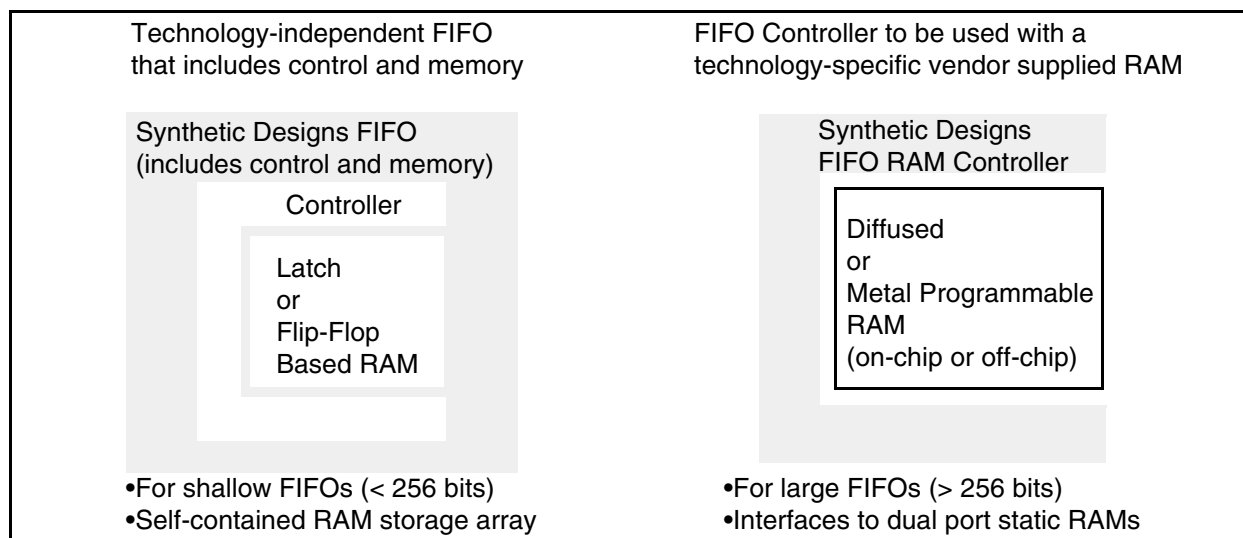
DWBB Memory – FIFO Overview

The FIFOs in this category address a broad array of design requirements. FIFOs, which include dual-port RAM memory arrays, are offered for both synchronous and asynchronous interfaces. The memory arrays are offered in two configurations: latch-based to minimize area, and D flip-flop-based to maximize testability. These two configurations also offer flexibility when working under design constraints, such as a requirement that no latches be employed. Flip-flop-based designs employ no clock gating to minimize skew and maximize performance. All FIFOs employ a FIFO RAM controller architecture in which there is no extended “fall-through” time required before reading contents just written.

Also offered are FIFO Controllers without the RAM array. They consist of control and flag logic and an interface to common ASIC dual port RAMs. Choosing between the two is typically based on the required size of the FIFO. For shallow FIFOs (less than 256 bits), synchronous or asynchronous FIFOs are available which include both memory and control in a single macro. These macros can be programmed via word width, depth, and level (almost-full flag) parameters.

For larger applications (greater than 256 bits), you can use the asynchronous FIFO Controller with a diffused or metal programmable RAM; see [Figure 1-1](#).

Figure 1-1 Memory: FIFOs and FIFO Controllers



All FIFOs and Controllers support full, empty, and programmable flag logic. Programmable flag logic may be statically or dynamically programmed. When statically programmed, the threshold comparison value is hardwired at synthesis compile time. When dynamically programmed, it may be changed during FIFO operation.

Related Topics

For a listing of Building Block components and associated datasheets, see:

- [DesignWare Building Block IP User Guide](#)

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